PRACTICAL – 4

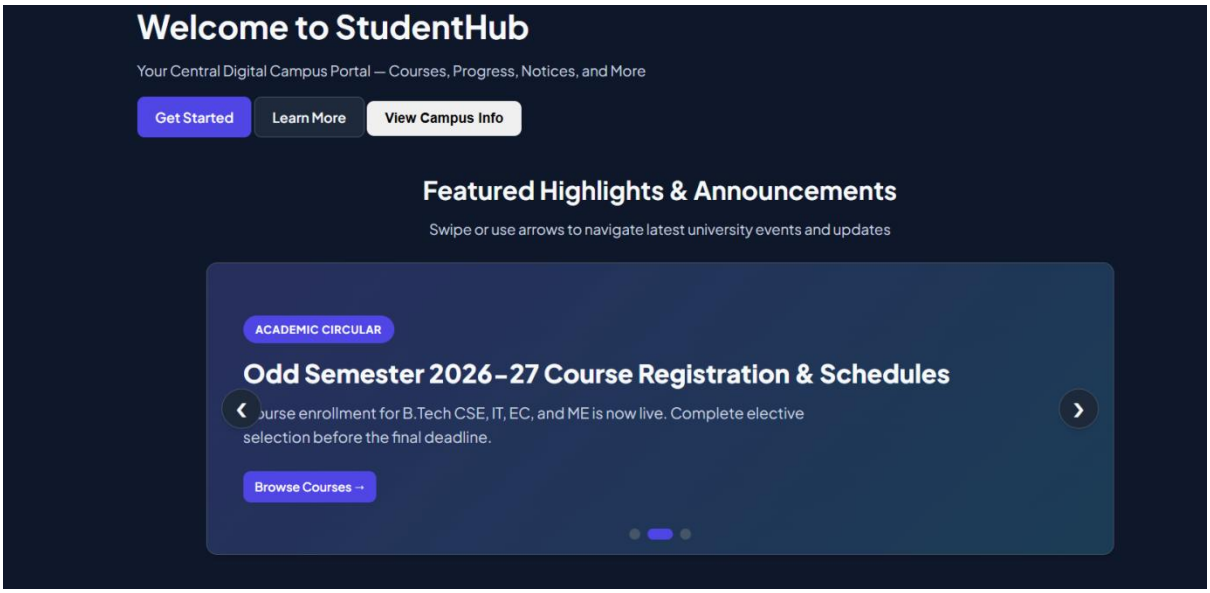
Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript.

The StudentHub website developed in the previous practical contains static HTML and CSS pages. In this practical, JavaScript is used to make the website interactive and user-friendly. The following dynamic components are implemented:

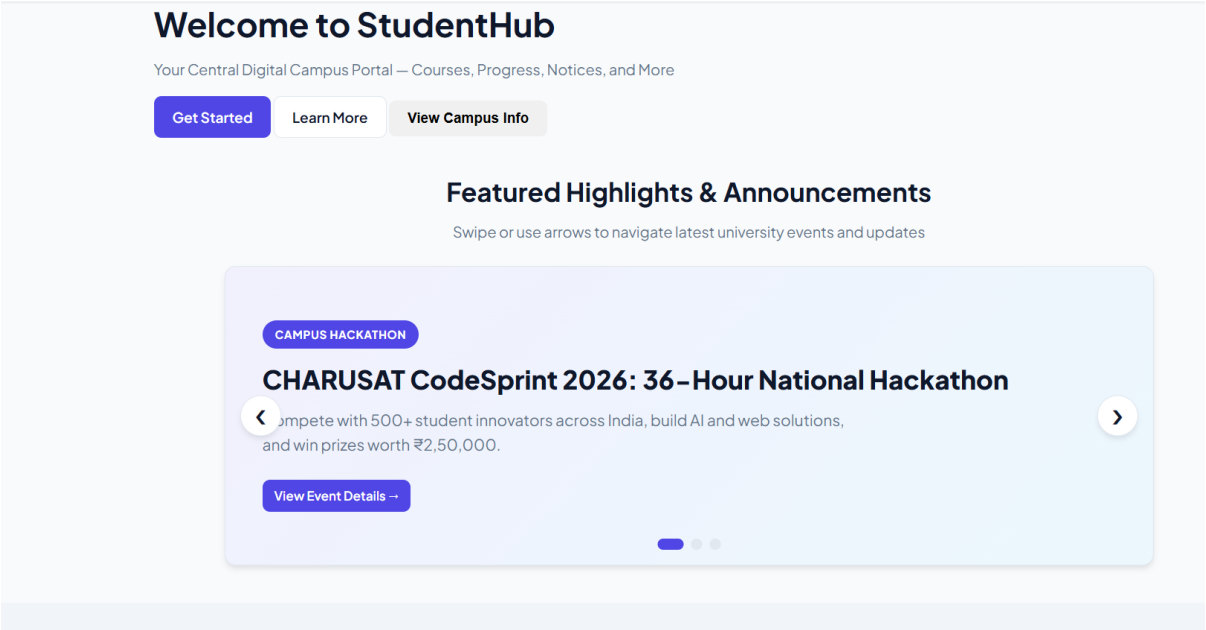
- Collapsible FAQ
- Modal popup
- Content slider
- Notification banner
- Hamburger menu
- Light/Dark theme switcher

Dark/Light Theme:

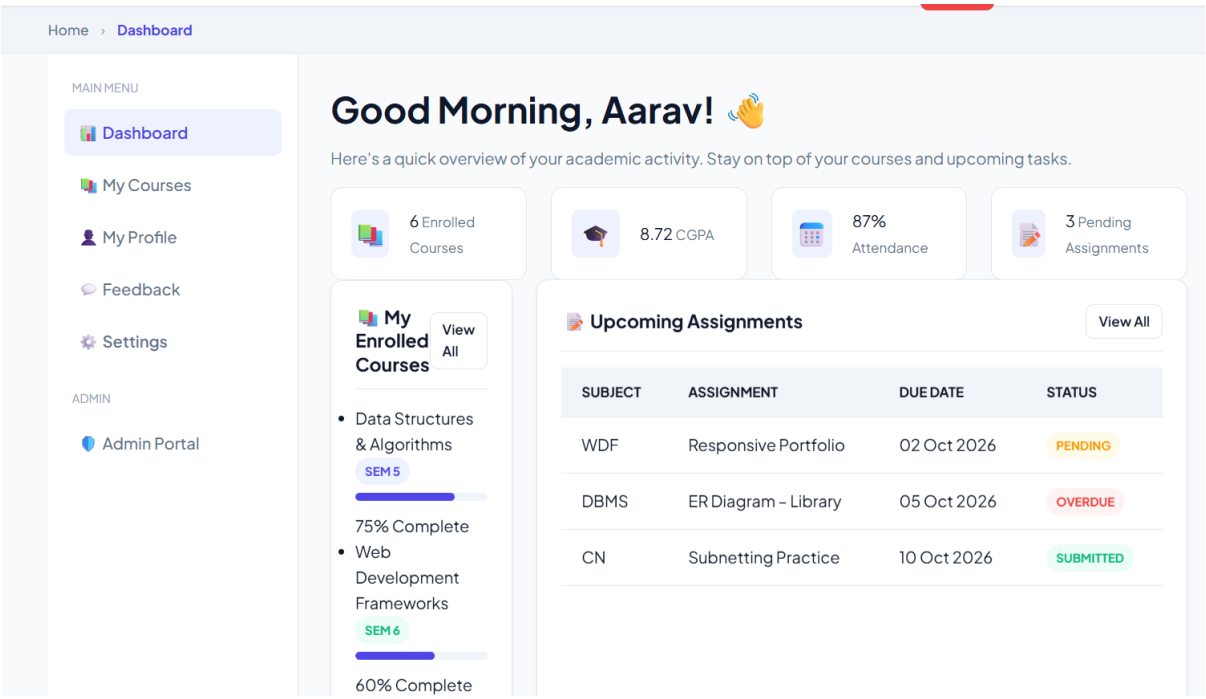
Dark Mode:

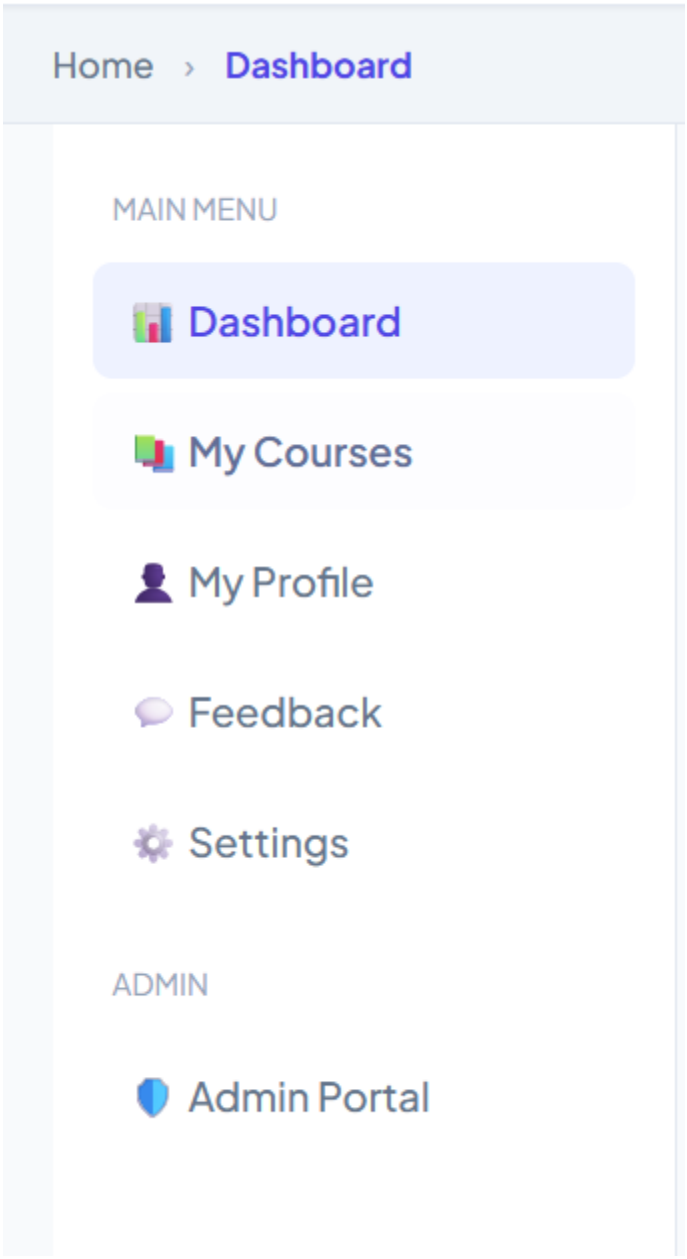


Light Mode:

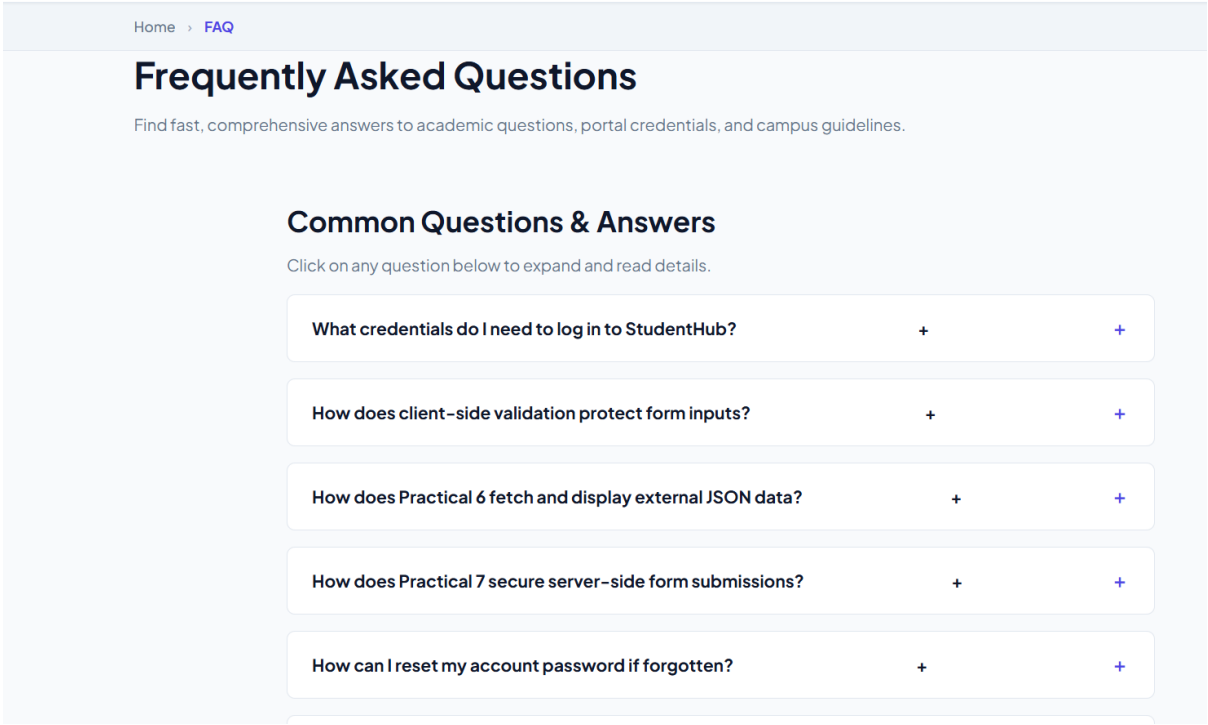


Hamburger Menu:

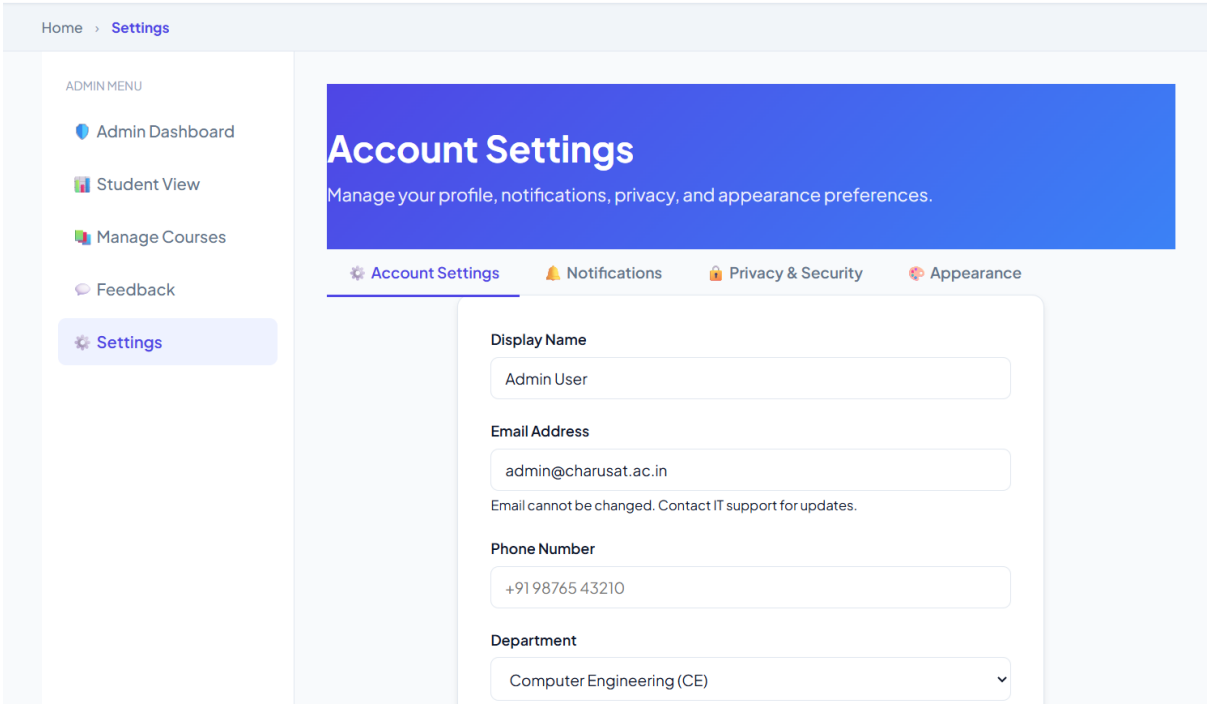



Home › Dashboard
MAIN MENU
Dashboard
My Courses
My Profile
Feedback
Settings
ADMIN
Admin Portal

Collapsible FAQ:



Content slider:



Key Questions:

1. How are DOM elements selected and modified?
Ans:DOM elements are selected using methods such as getElementById() and querySelector(). They can be modified using properties and methods such as style, textContent, and classList.

2. Are event listeners attached correctly?
Ans: Yes. addEventListener() is used to respond to events such as button clicks.

3. Is localStorage used for remembering the theme?
Ans: Yes. The selected theme is stored in localStorage, so it can be restored when the page is opened again.

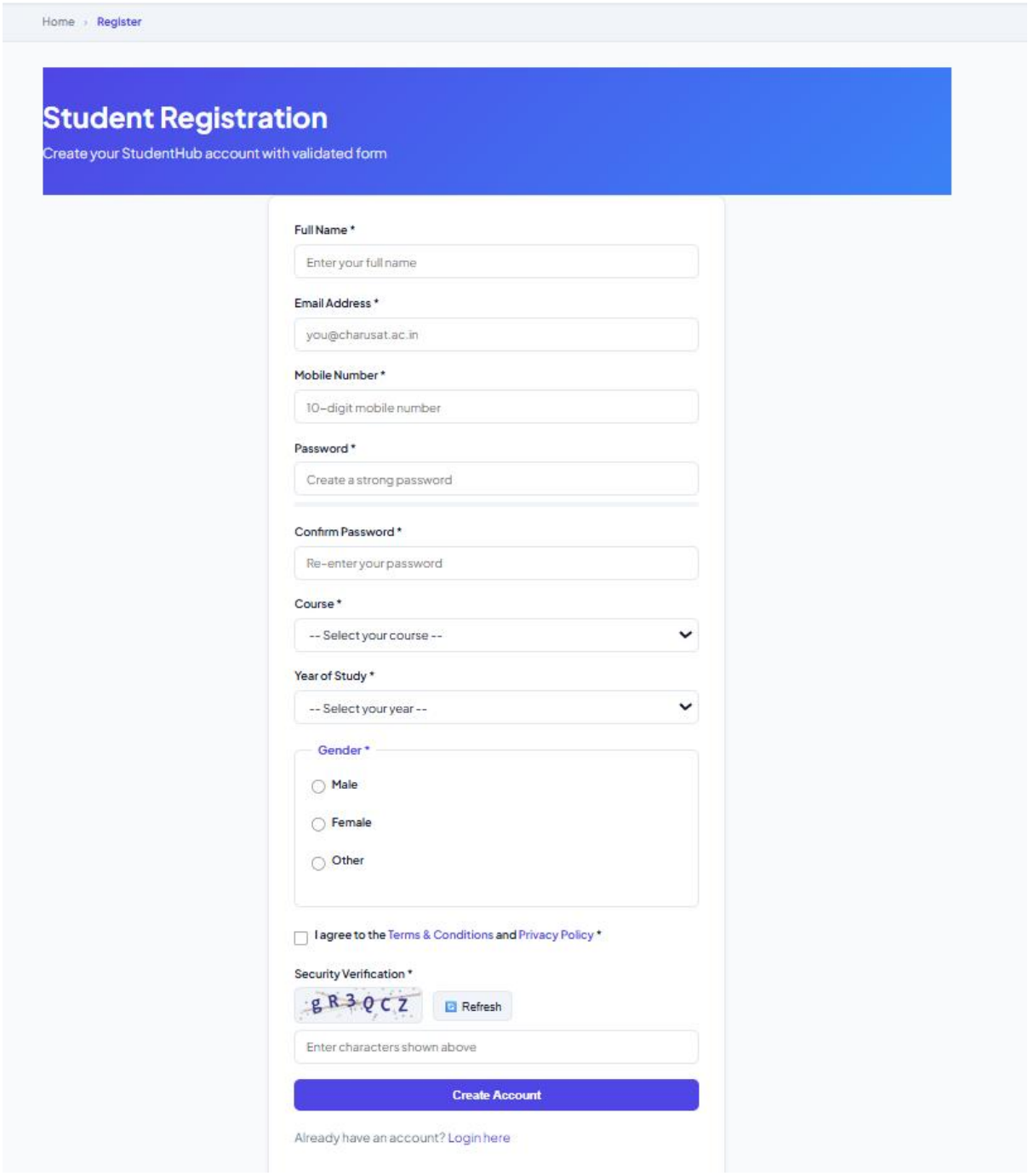
4. Does interactivity improve usability without breaking accessibility?
Ans: Yes. Buttons are used for interactive actions, and accessibility attributes such as labels can be added so keyboard and assistive-technology users can understand the controls.

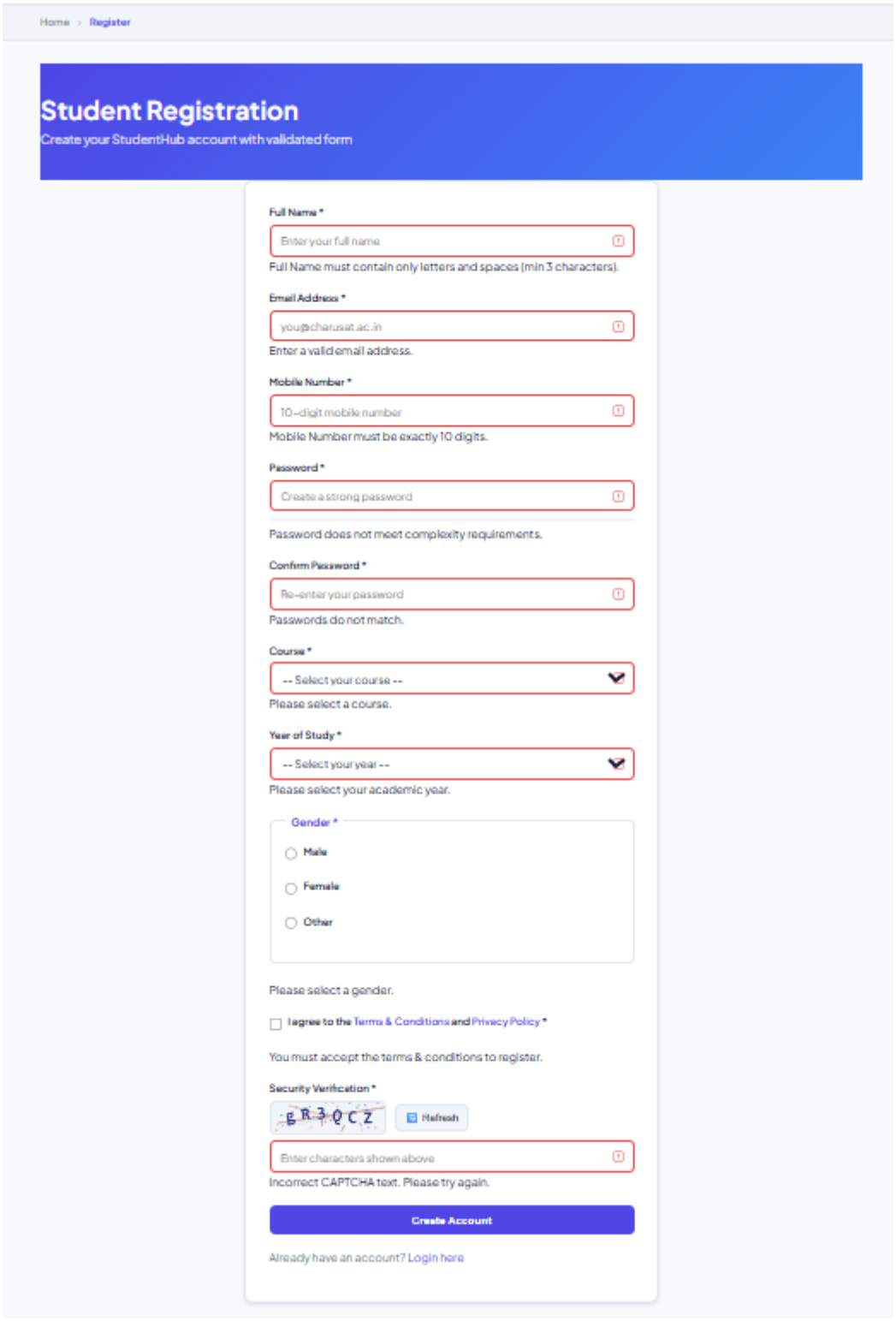
GitHub Link: https://github.com/smit-2406/studenthub-website/tree/main/WD

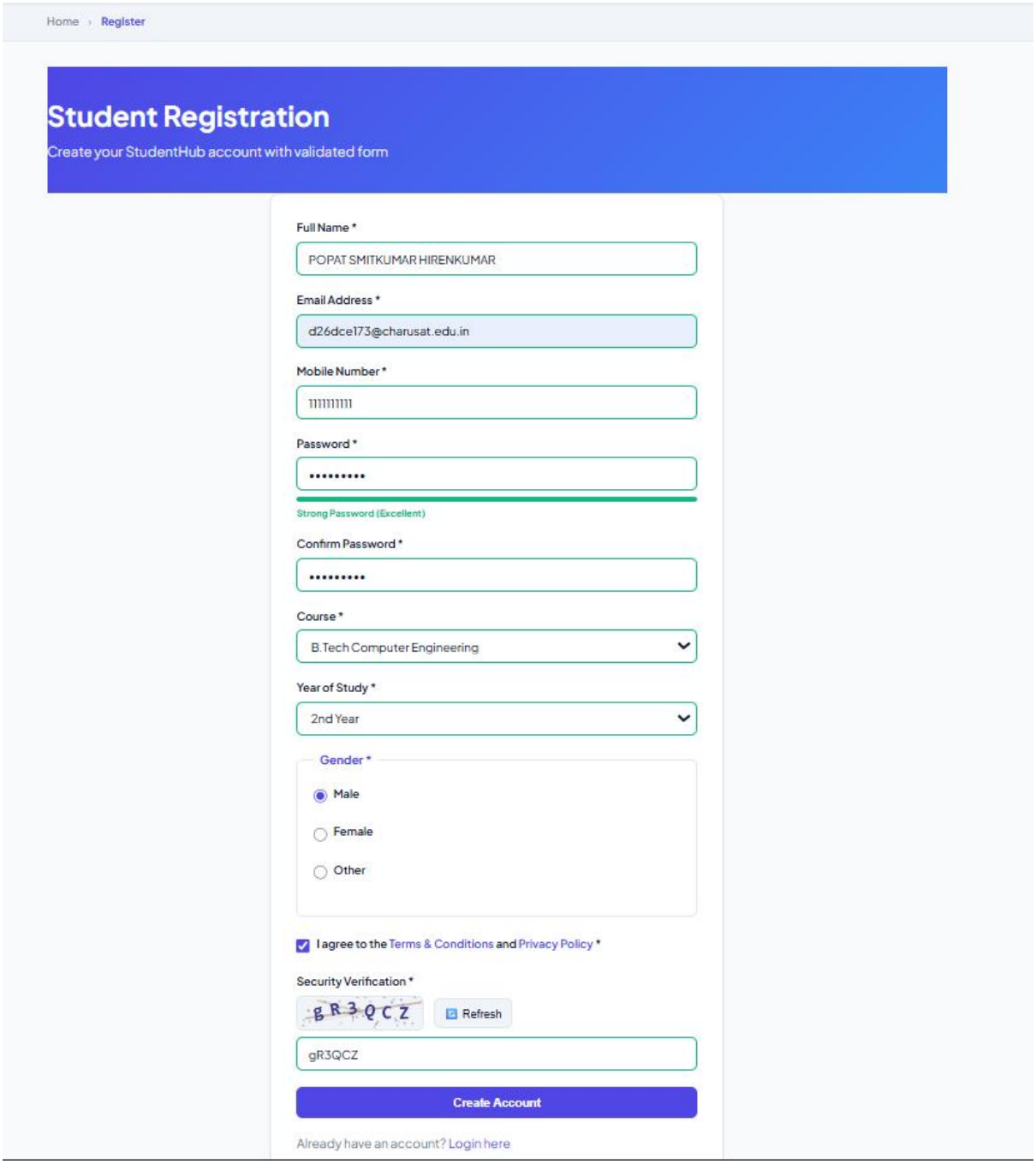
PRACTICAL – 5

Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

Create a student registration form for the StudentHub website with proper HTML5 input types and JavaScript-based client-side validation. The form should validate all required fields, display errors near the relevant fields, check password strength, and provide an accessible and user-friendly registration experience.


Home › Register
Student Registration
Create your StudentHub account with validated form
Full Name *
Enter your full name
Email Address *
you@charusat.ac.in
Mobile Number *
10-digit mobile number
Password *
Create a strong password
Confirm Password *
Re-enter your password
Course *
-- Select your course --
Year of Study *
-- Select your year --
Gender *
Male
Female
Other
I agree to the Terms & Conditions and Privacy Policy *
Security Verification *
gR3QCZ
Refresh
Enter characters shown above
Create Account
Already have an account? Login here


Home › Register
Student Registration
Create your StudentHub account with validated form
Full Name *
Enter your full name
Full Name must contain only letters and spaces (min 3 characters).
Email Address *
you@charusat.ac.in
Enter a valid email address.
Mobile Number *
10-digit mobile number
Mobile Number must be exactly 10 digits.
Password *
Create a strong password
Password does not meet complexity requirements.
Confirm Password *
Re-enter your password
Passwords do not match.
Course *
-- Select your course --
Please select a course.
Year of Study *
-- Select your year --
Please select your academic year.
Gender *
Male
Female
Other
Please select a gender.
I agree to the Terms & Conditions and Privacy Policy *
You must accept the terms & conditions to register.
Security Verification *
gR3QCZ
Refresh
Enter characters shown above
Incorrect CAPTCHA text. Please try again.
Create Account
Already have an account? Login here


Home › Register
Student Registration
Create your StudentHub account with validated form
Full Name *
POPAT SMITKUMAR HIRENKUMAR
Email Address *
d26dce173@charusat.edu.in
Mobile Number *
11111111
Password *
Strong Password (Excellent)
Confirm Password *
Course *
B.Tech Computer Engineering
Year of Study *
2nd Year
Gender *
Male
Female
Other
I agree to the Terms & Conditions and Privacy Policy *
Security Verification *
gR3QCZ
Refresh
gR3QCZ
Create Account
Already have an account? Login here

Key Questions:

1. Are correct input types and attributes used?

Ans: Yes. Appropriate HTML5 input types such as email, tel, password, radio, and checkbox are used. Attributes such as required, autocomplete, maxlength, and aria-describedby improve validation, usability, and accessibility.

Q2. Are validation errors displayed near the relevant fields?

Ans: Yes. Each input has a dedicated error-message area immediately below it. This makes it easier for the user to understand and correct the problem.

3. Is password strength checked?

Ans: Yes. A password-strength indicator is provided. JavaScript checks the password length and the presence of uppercase letters, lowercase letters, numbers, and special characters.

4. Is the form accessible using keyboard and screen-reader-friendly labels?

Ans: Yes. Every input has an associated <label>. aria-describedby and aria-live are used for error messages, allowing screen readers to communicate validation errors to users.

GitHub Link: https://github.com/smit-2406/studenthub-website/tree/main/WD

PRACTICAL – 6

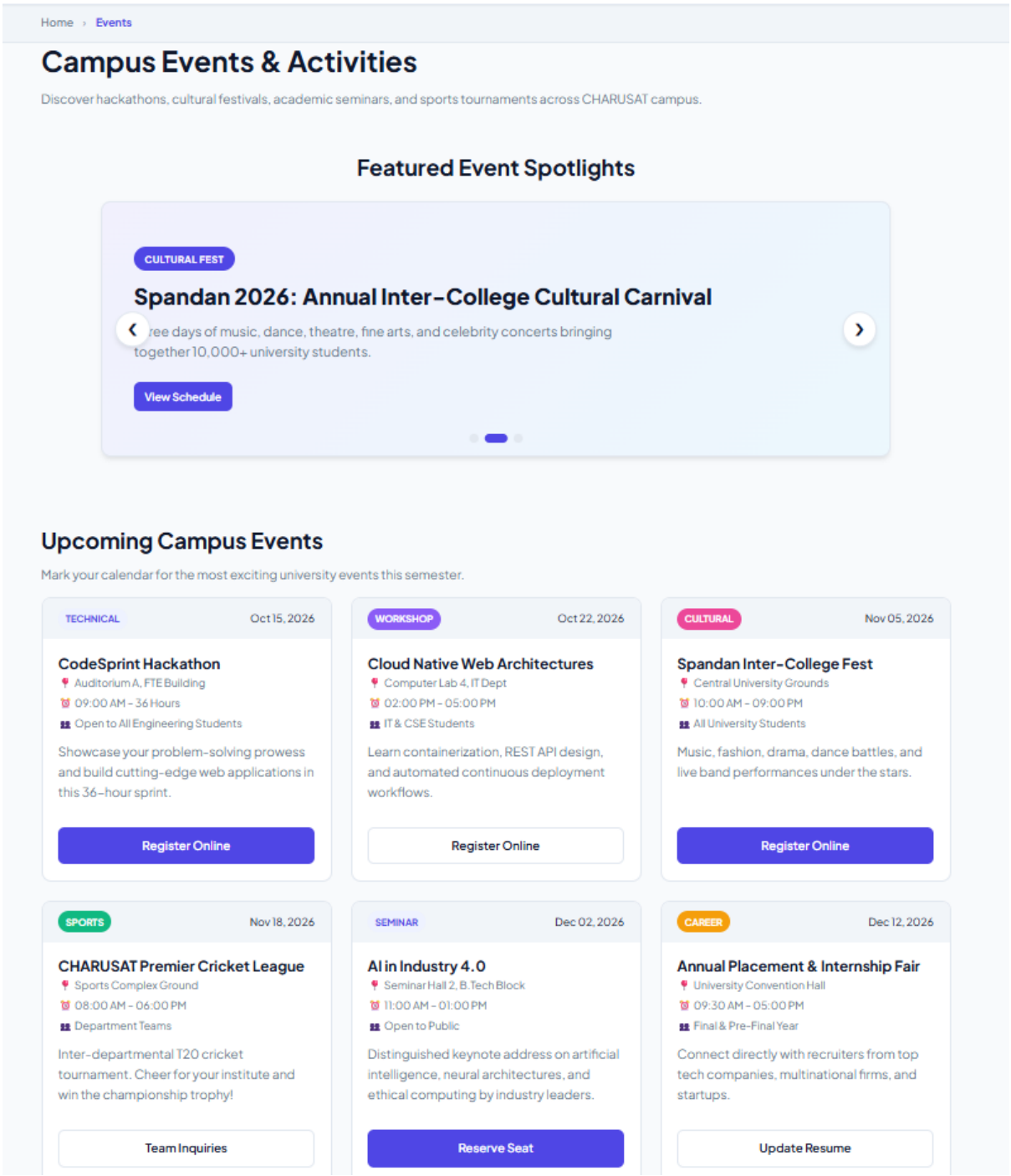
Display event lists, student profiles, notices, or FAQs from external JSON files using Fetch API. Implement dynamic rendering, search, filter, sorting, and pagination.

JSON (JavaScript Object Notation) is a lightweight data format used to store and exchange structured data. In StudentHub Practical 6, JSON files are used as external data sources for information such as events, student profiles, notices, and FAQs.

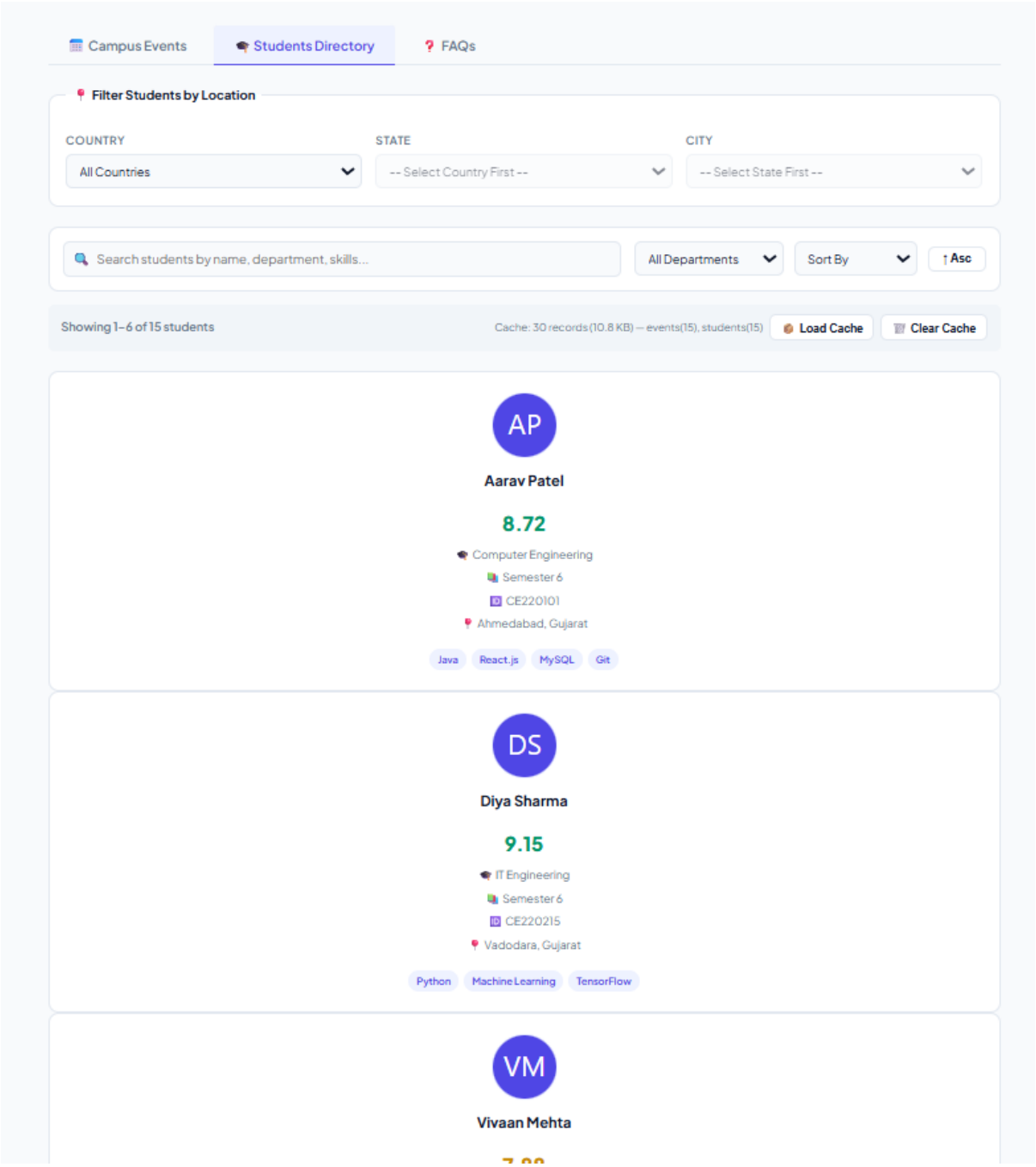
The JSON data is retrieved using the JavaScript Fetch API. After fetching the file, the response is converted into a JavaScript object using the response.json() method. The data is then dynamically displayed on the webpage.

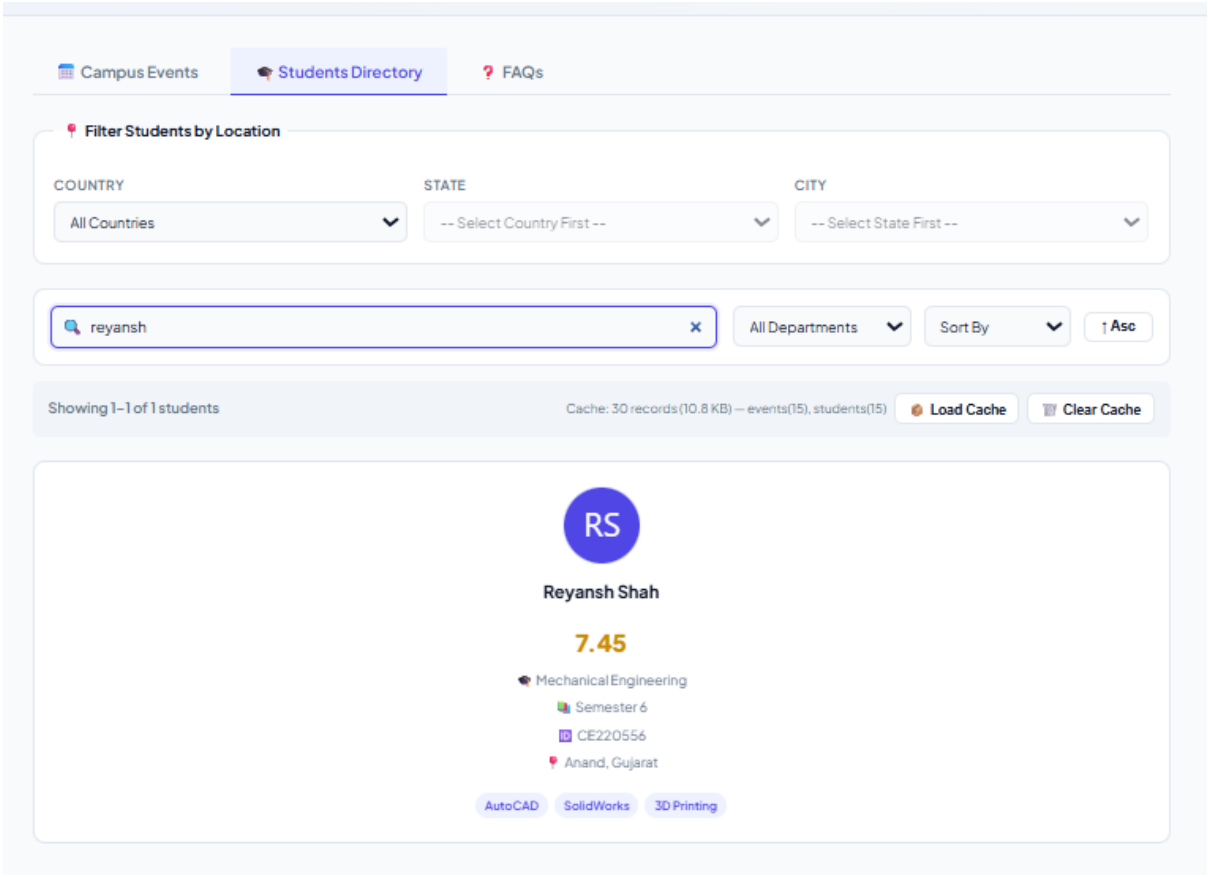
JavaScript array methods such as filter(), sort(), and slice() are used to implement searching, filtering, sorting, and pagination.

Event.html:

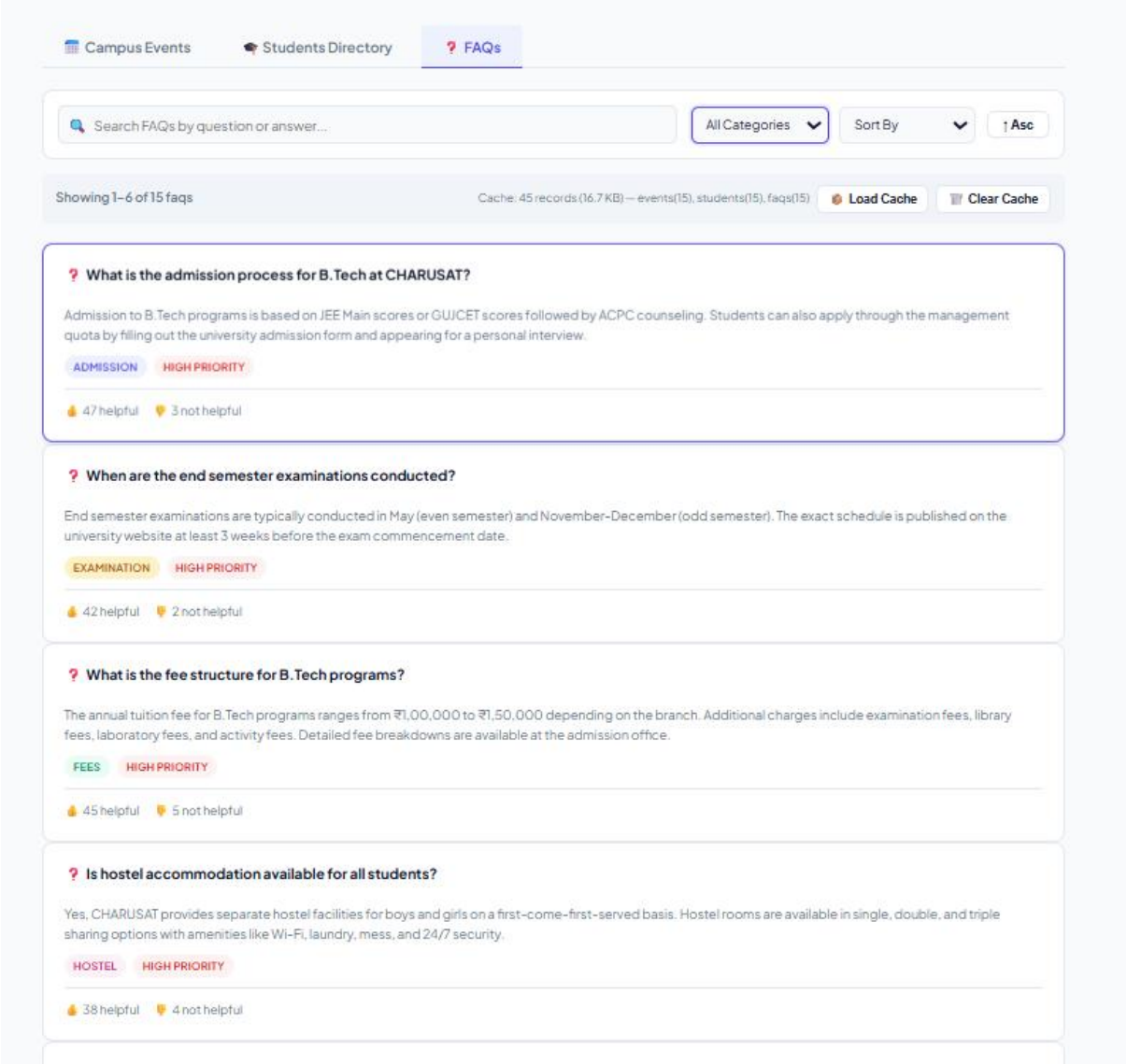

Home › Events
Campus Events & Activities
Discover hackathons, cultural festivals, academic seminars, and sports tournaments across CHARUSAT campus.
Featured Event Spotlights
CULTURAL FEST
Spandan 2026: Annual Inter-College Cultural Carnival
ree days of music, dance, theatre, fine arts, and celebrity concerts bringing together 10,000+ university students.
View Schedule
Upcoming Campus Events
Mark your calendar for the most exciting university events this semester.
TECHNICAL
Oct 15, 2026
CodeSprint Hackathon
Auditorium A, FTE Building
09:00 AM - 36 Hours
Open to All Engineering Students
Showcase your problem-solving prowess and build cutting-edge web applications in this 36-hour sprint.
Register Online
WORKSHOP
Oct 22, 2026
Cloud Native Web Architectures
Computer Lab 4, IT Dept
02:00 PM - 05:00 PM
IT & CSE Students
Learn containerization, REST API design, and automated continuous deployment workflows.
Register Online
CULTURAL
Nov 05, 2026
Spandan Inter-College Fest
Central University Grounds
10:00 AM - 09:00 PM
All University Students
Music, fashion, drama, dance battles, and live band performances under the stars.
Register Online
SPORTS
Nov 18, 2026
CHARUSAT Premier Cricket League
Sports Complex Ground
08:00 AM - 06:00 PM
Department Teams
Inter-departmental T20 cricket tournament. Cheer for your institute and win the championship trophy!
Team Inquiries
SEMINAR
Dec 02, 2026
AI in Industry 4.0
Seminar Hall 2, B.Tech Block
11:00 AM - 01:00 PM
Open to Public
Distinguished keynote address on artificial intelligence, neural architectures, and ethical computing by industry leaders.
Reserve Seat
CAREER
Dec 12, 2026
Annual Placement & Internship Fair
University Convention Hall
09:30 AM - 05:00 PM
Final & Pre-Final Year
Connect directly with recruiters from top tech companies, multinational firms, and startups.
Update Resume

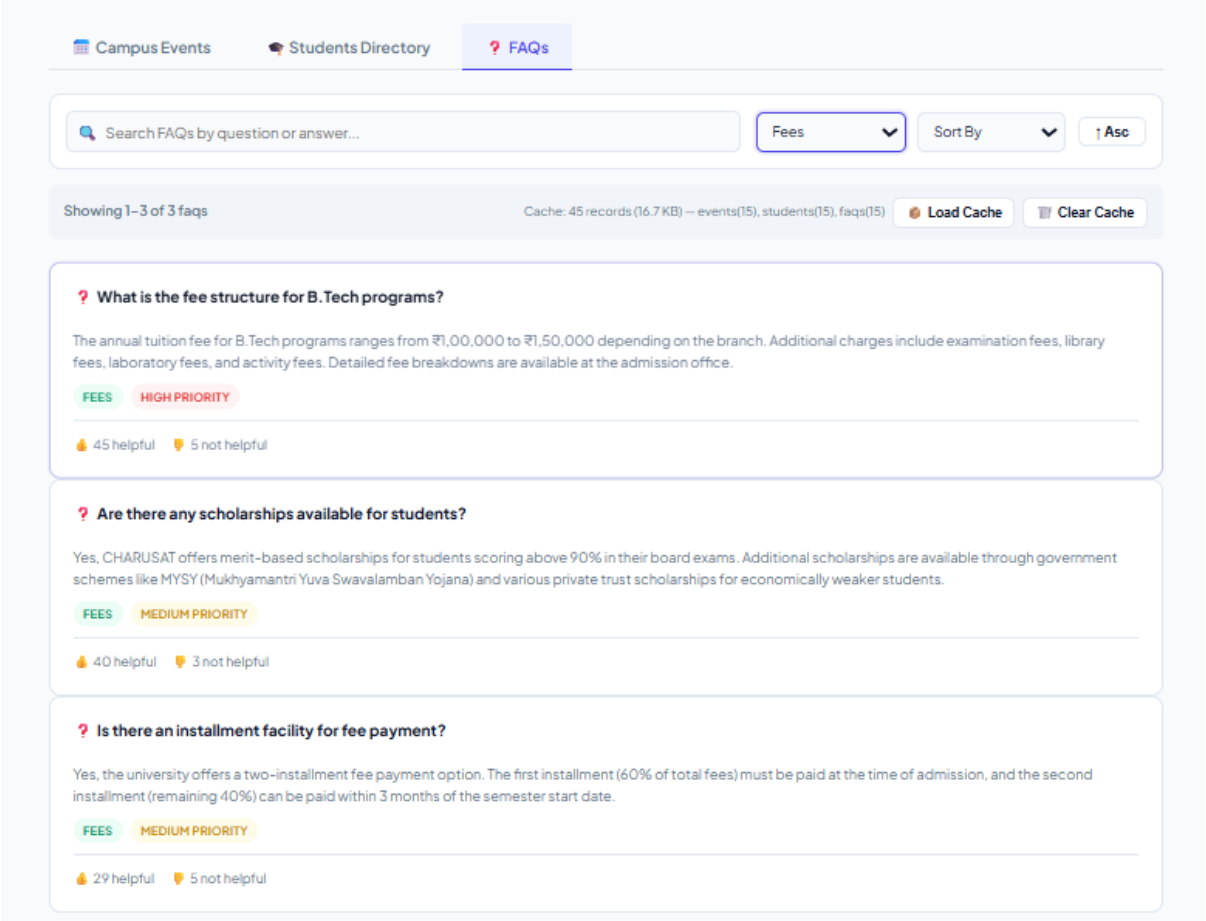
Profile.html:




Campus Events
Students Directory
FAQs
Filter Students by Location
COUNTRY
All Countries
STATE
-- Select Country First --
CITY
-- Select State First --
reyansh
All Departments
Sort By
↑ Asc
Showing 1–1 of 1 students
Cache: 30 records (10.8 KB) — events(15), students(15)
Load Cache
Clear Cache
RS
Reyansh Shah
7.45
Mechanical Engineering
Semester 6
CE220556
Anand, Gujarat
AutoCAD
SolidWorks
3D Printing

Faq.html:


Campus Events
Students Directory
FAQs
Search FAQs by question or answer...
All Categories
Sort By
↑ Asc
Showing 1–6 of 15 faqs
Cache: 45 records (16.7 KB) — events(15), students(15), faqs(15)
Load Cache
Clear Cache
What is the admission process for B.Tech at CHARUSAT?
Admission to B.Tech programs is based on JEE Main scores or GUJCET scores followed by ACPC counseling. Students can also apply through the management quota by filling out the university admission form and appearing for a personal interview.
ADMISSION
HIGH PRIORITY
47 helpful
3 not helpful
When are the end semester examinations conducted?
End semester examinations are typically conducted in May (even semester) and November-December (odd semester). The exact schedule is published on the university website at least 3 weeks before the exam commencement date.
EXAMINATION
HIGH PRIORITY
42 helpful
2 not helpful
What is the fee structure for B.Tech programs?
The annual tuition fee for B.Tech programs ranges from ₹1,00,000 to ₹1,50,000 depending on the branch. Additional charges include examination fees, library fees, laboratory fees, and activity fees. Detailed fee breakdowns are available at the admission office.
FEES
HIGH PRIORITY
45 helpful
5 not helpful
Is hostel accommodation available for all students?
Yes, CHARUSAT provides separate hostel facilities for boys and girls on a first-come-first-served basis. Hostel rooms are available in single, double, and triple sharing options with amenities like Wi-Fi, laundry, mess, and 24/7 security.
HOSTEL
HIGH PRIORITY
38 helpful
4 not helpful



Key Questions:

1. How is JSON fetched, parsed, and rendered?

Ans: Yes. Appropriate HTML5 input types such as email, tel, password, radio, and checkbox are used. Attributes such as required, autocomplete, maxlength, and aria-describedby improve validation, usability, and accessibility.

Q2. Which array methods are used for search, filter, sort, and pagination?

Ans: Yes. Each input has a dedicated error-message area immediately below it. This makes it easier for the user to understand and correct the problem.

3. How are loading and error states handled?

Ans: Yes. A password-strength indicator is provided. JavaScript checks the password length and the presence of uppercase letters, lowercase letters, numbers, and special characters.

4. How is modularity maintained in JavaScript files?

Ans: Yes. Every input has an associated <label>. aria-describedby and aria-live are used for error messages, allowing screen readers to communicate validation errors to users.

GitHub Link: https://github.com/smit-2406/studenthub-website/tree/main/WD

PRACTICAL – 7

Process submitted registration/contact form data using PHP. Validate and sanitize inputs on the server side and store records in CSV or JSON file format. Display success/error messages.

Register.php:

Code:

register.php

```php
<?php
$message = "";
$messageType = "";

if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $name = trim($_POST["name"] ?? "");
    $email = trim($_POST["email"] ?? "");
    $mobile = trim($_POST["mobile"] ?? "");
    $course = trim($_POST["course"] ?? "");

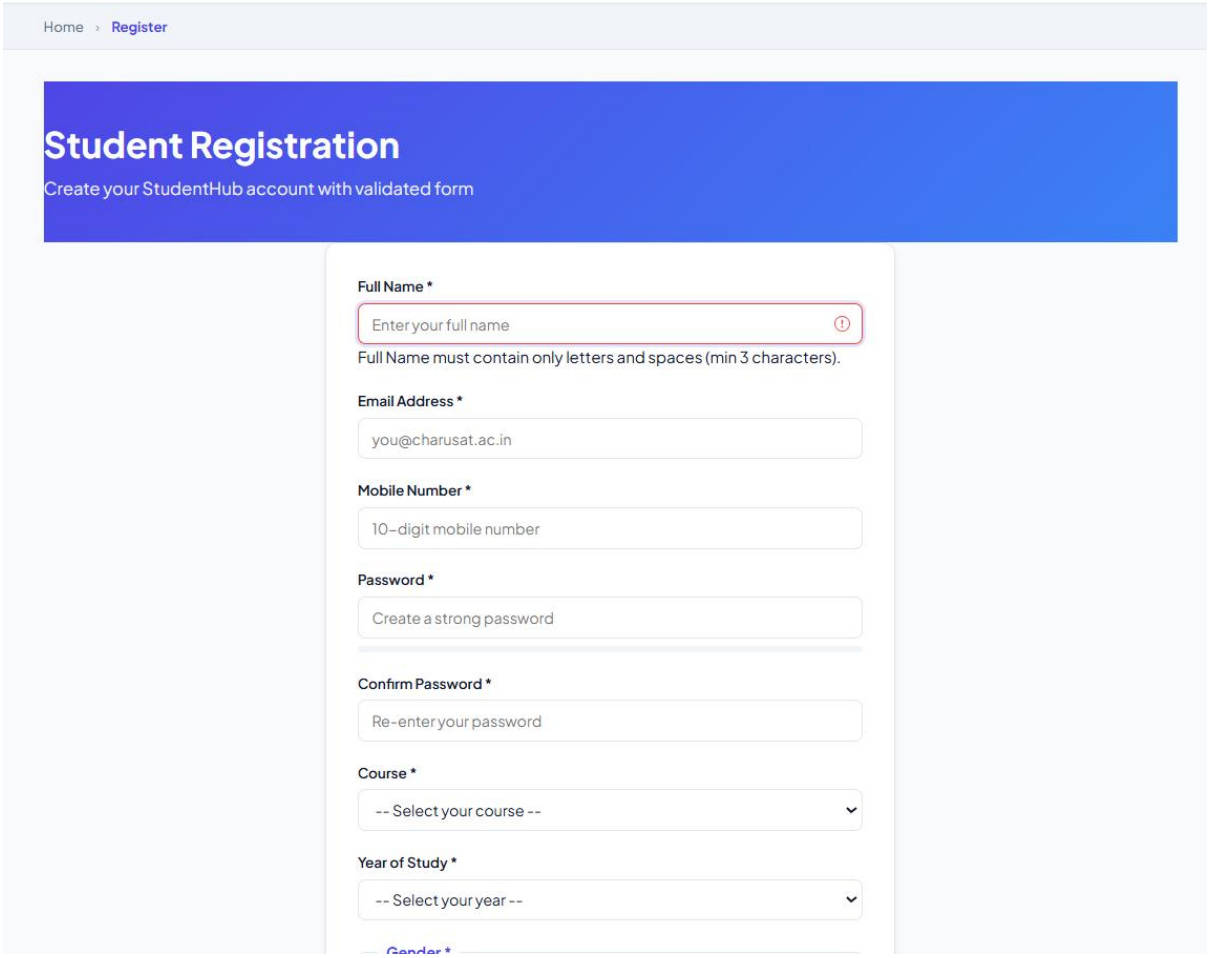
    if ($name === "" || $email === "" || $mobile === "" || $course === "") {
        $message = "Please fill all required fields.";
        $messageType = "error";
    } elseif (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $message = "Please enter a valid email address.";
        $messageType = "error";
    } elseif (!preg_match("/^[0-9]{10}$/", $mobile)) {
        $message = "Mobile number must contain exactly 10 digits.";
        $messageType = "error";
    } else {
        $name = htmlspecialchars($name, ENT_QUOTES, "UTF-8");
        $email = htmlspecialchars($email, ENT_QUOTES, "UTF-8");
        $mobile = htmlspecialchars($mobile, ENT_QUOTES, "UTF-8");
        $course = htmlspecialchars($course, ENT_QUOTES, "UTF-8");

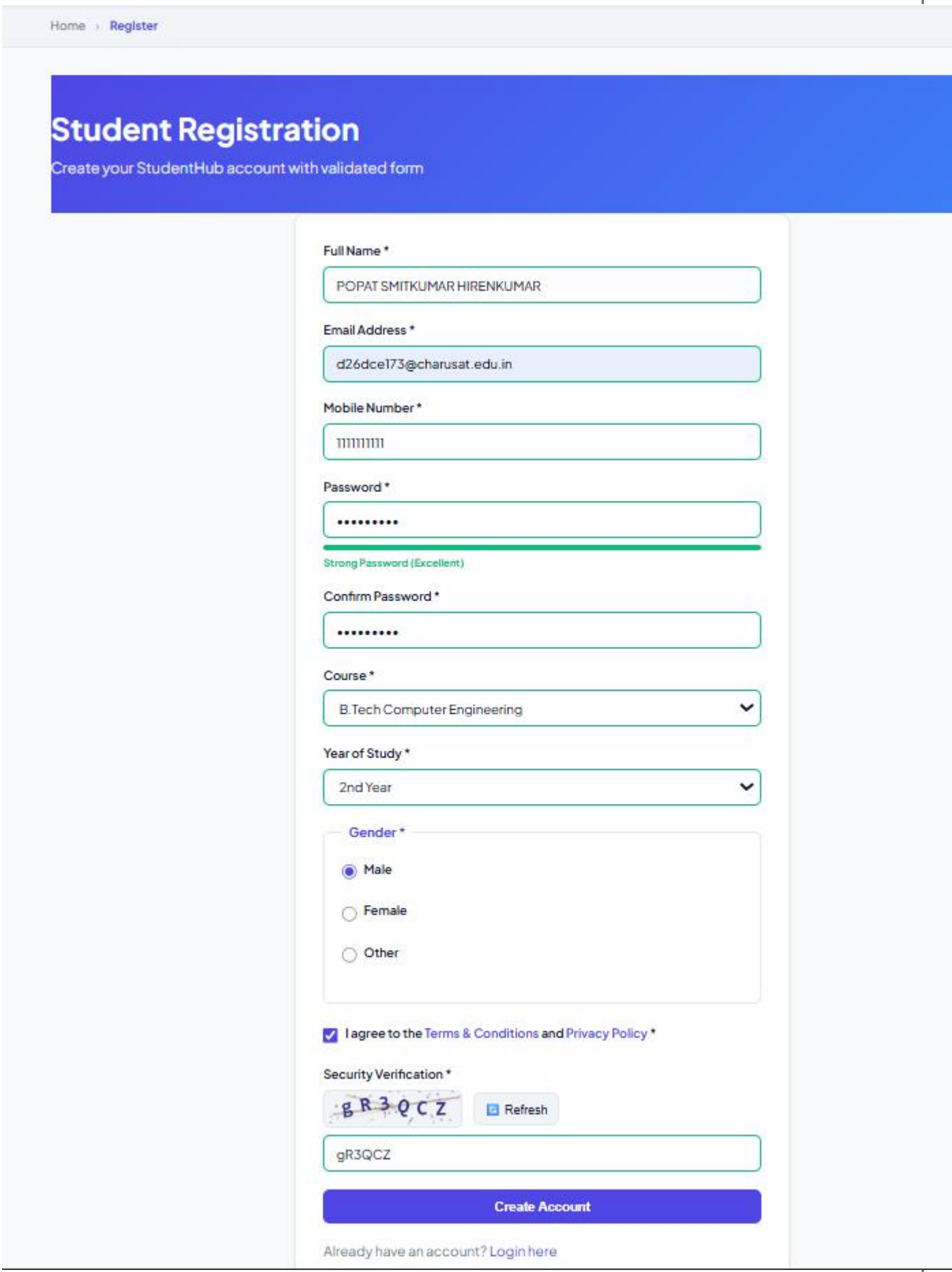
        $file = __DIR__ . "/data/registrations.csv";
        if (!file_exists($file)) {
            $fp = fopen($file, "w");
            fputcsv($fp, ["Name", "Email", "Mobile", "Course"]);
            fclose($fp);
        }

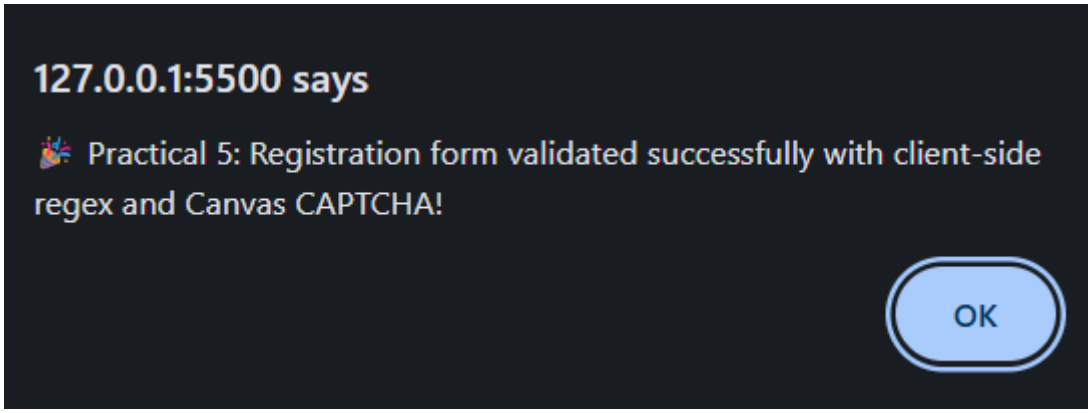
        $fp = fopen($file, "a");
        if ($fp) {
            fputcsv($fp, [$name, $email, $mobile, $course]);
            fclose($fp);
            $message = "Registration successful!";
            $messageType = "success";
        } else {
            $message = "Unable to save registration.";
            $messageType = "error";
        }
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>StudentHub - Registration</title>
<link rel="stylesheet" href="styles.css">
<style>
.form-box{max-width:600px;padding:25px;background:#fff;border-radius:10px}
.form-group{margin-bottom:18px}.form-group label{display:block;margin-bottom:6px;font-weight:bold}
.form-group input,.form-group select{width:100%;padding:10px;box-sizing:border-box}
.form-box button{padding:10px 20px;cursor:pointer}
.success{padding:12px;margin-bottom:15px;background:#d4edda;color:#155724}
.error{padding:12px;margin-bottom:15px;background:#f8d7da;color:#721c24}
</style>
</head>
<body>
<div class="layout">
<aside class="sidebar">
<header><h2>STUDENTHUB</h2><p>Student Portal</p></header>
<nav aria-label="Student Portal Navigation">
<ul>
<li><a href="dashboard.html">Dashboard</a></li>
<li><a href="announcements.html">Announcements</a></li>
<li><a href="timetable.html">Courses &amp;amp; Timetable</a></li>
<li><a href="attendance.html">Attendance</a></li>
<li><a href="results.html">Results</a></li>
<li><a href="materials.html">Study Resources</a></li>
<li><a href="events.html">Events</a></li>
<li><a href="contact.php">Support</a></li>
<li><a href="register.php" aria-current="page">Register</a></li>
</ul>
</nav>
</aside>
<main class="content">
<nav aria-label="Breadcrumb"><ol><li><a href="dashboard.html">Home</a></li><li aria-current="page">Register</li></ol></nav>
<header><p>ACCOUNT</p><h1>Student Registration</h1><p>Create your StudentHub account.</p></header>
<section class="form-box">
<?php if ($message !== ""): ?><div class="<?php echo $messageType; ?>"><?php echo $message; ?></div><?php endif; ?>
<form method="POST" action="register.php">
<div class="form-group"><label for="name">Full Name</label><input type="text" id="name" name="name" required></div>
<div class="form-group"><label for="email">Email</label><input type="email" id="email" name="email" required></div>
<div class="form-group"><label for="mobile">Mobile Number</label><input type="text" id="mobile" name="mobile" maxlength="10" required></div>
<div class="form-group"><label for="course">Course</label>
<select id="course" name="course" required>
<option value="">Select Course</option><option>Computer Engineering</option><option>Information Technology</option><option>Computer Science</option>
</select></div>
<button type="submit">Register</button>
</form>
</section>
</main>
</div>
</body>
</html>
```

Output:


Home › Register
Student Registration
Create your StudentHub account with validated form
Full Name *
Enter your full name
Full Name must contain only letters and spaces (min 3 characters).
Email Address *
you@charusat.ac.in
Mobile Number *
10-digit mobile number
Password *
Create a strong password
Confirm Password *
Re-enter your password
Course *
-- Select your course --
Year of Study *
-- Select your year --


Home › Register
Student Registration
Create your StudentHub account with validated form
Full Name *
POPAT SMITKUMAR HIRENKUMAR
Email Address *
d26dce173@charusat.edu.in
Mobile Number *
Password *
Strong Password (Excellent)
Confirm Password *
Course *
B.Tech Computer Engineering
Year of Study *
2nd Year
Gender *
Male
Female
Other
I agree to the Terms & Conditions and Privacy Policy *
Security Verification *
gR3QCZ
Refresh
gR3QCZ
Create Account
Already have an account? Login here


127.0.0.1:5500 says
Practical 5: Registration form validated successfully with client-side regex and Canvas CAPTCHA!
OK

Contact.php:

Code:

contact.php

```php
<?php
$message = "";
$messageType = "";

if ($_SERVER["REQUEST_METHOD"] === "POST") {
    $subject = trim($_POST["subject"] ?? "");
    $category = trim($_POST["category"] ?? "");
    $details = trim($_POST["details"] ?? "");

    if ($subject === "" || $category === "" || $details === "") {
        $message = "All fields are required.";
        $messageType = "error";
    } else {
        $subject = htmlspecialchars($subject, ENT_QUOTES, "UTF-8");
        $category = htmlspecialchars($category, ENT_QUOTES, "UTF-8");
        $details = htmlspecialchars($details, ENT_QUOTES, "UTF-8");

        $file = __DIR__ . "/data/contacts.json";
        $contacts = [];

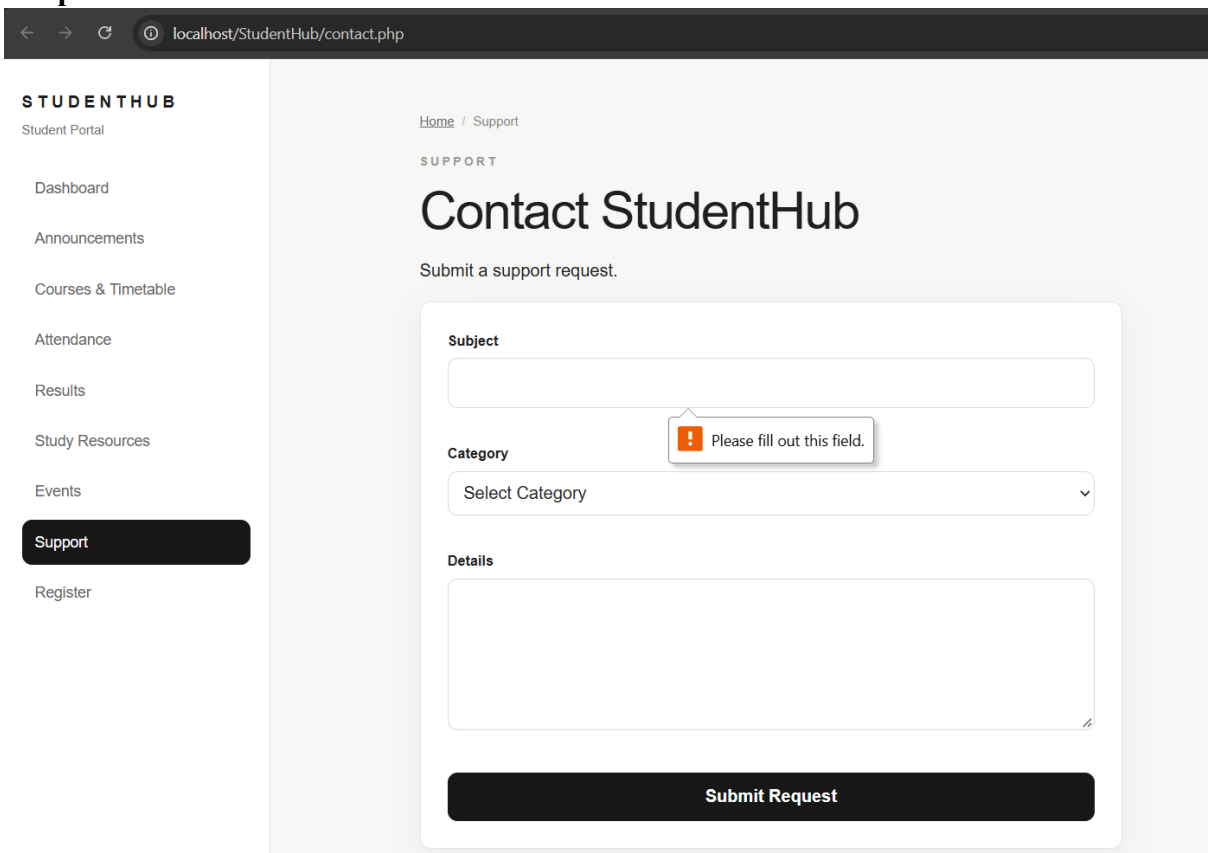
        if (file_exists($file)) {
            $json = file_get_contents($file);
            $decoded = json_decode($json, true);
            if (is_array($decoded)) $contacts = $decoded;
        }

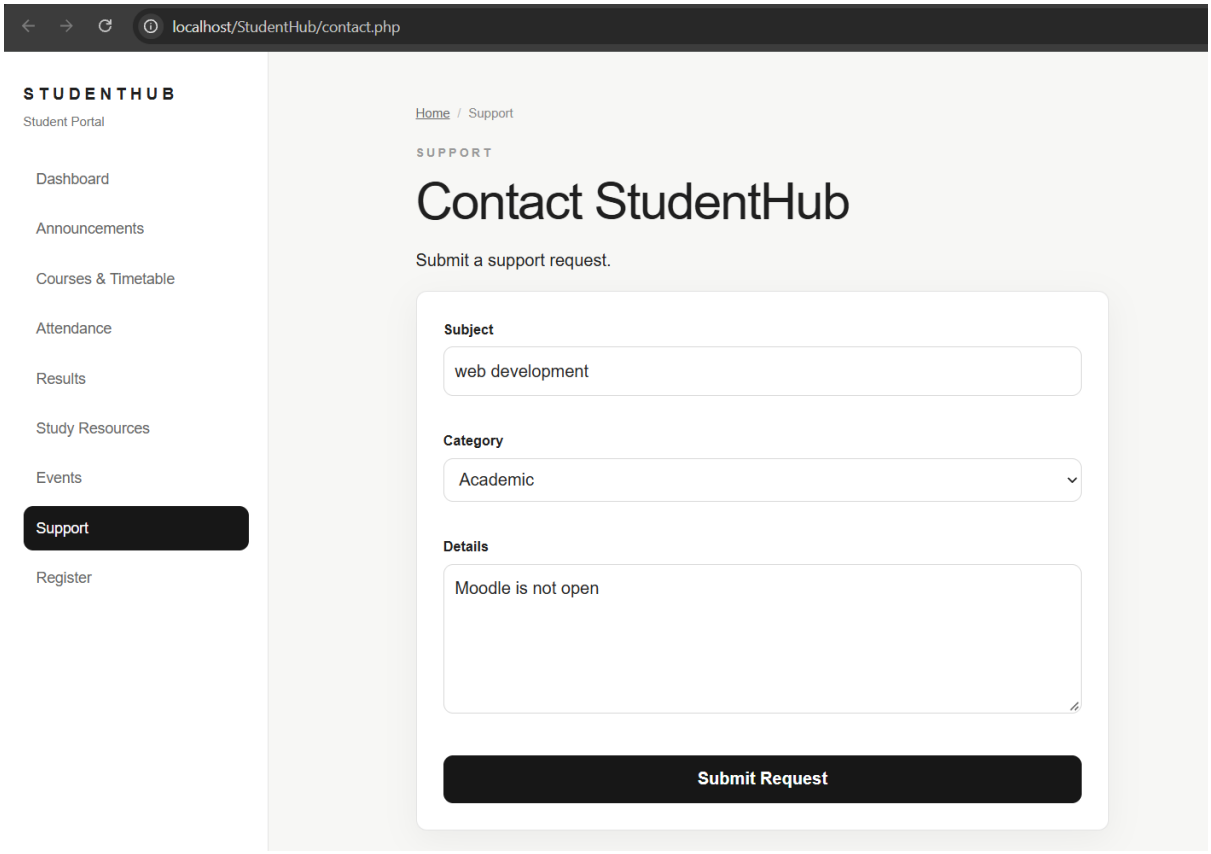
        $contacts[] = [
            "subject" => $subject,
            "category" => $category,
            "details" => $details,
            "date" => date("Y-m-d H:i:s")
        ];

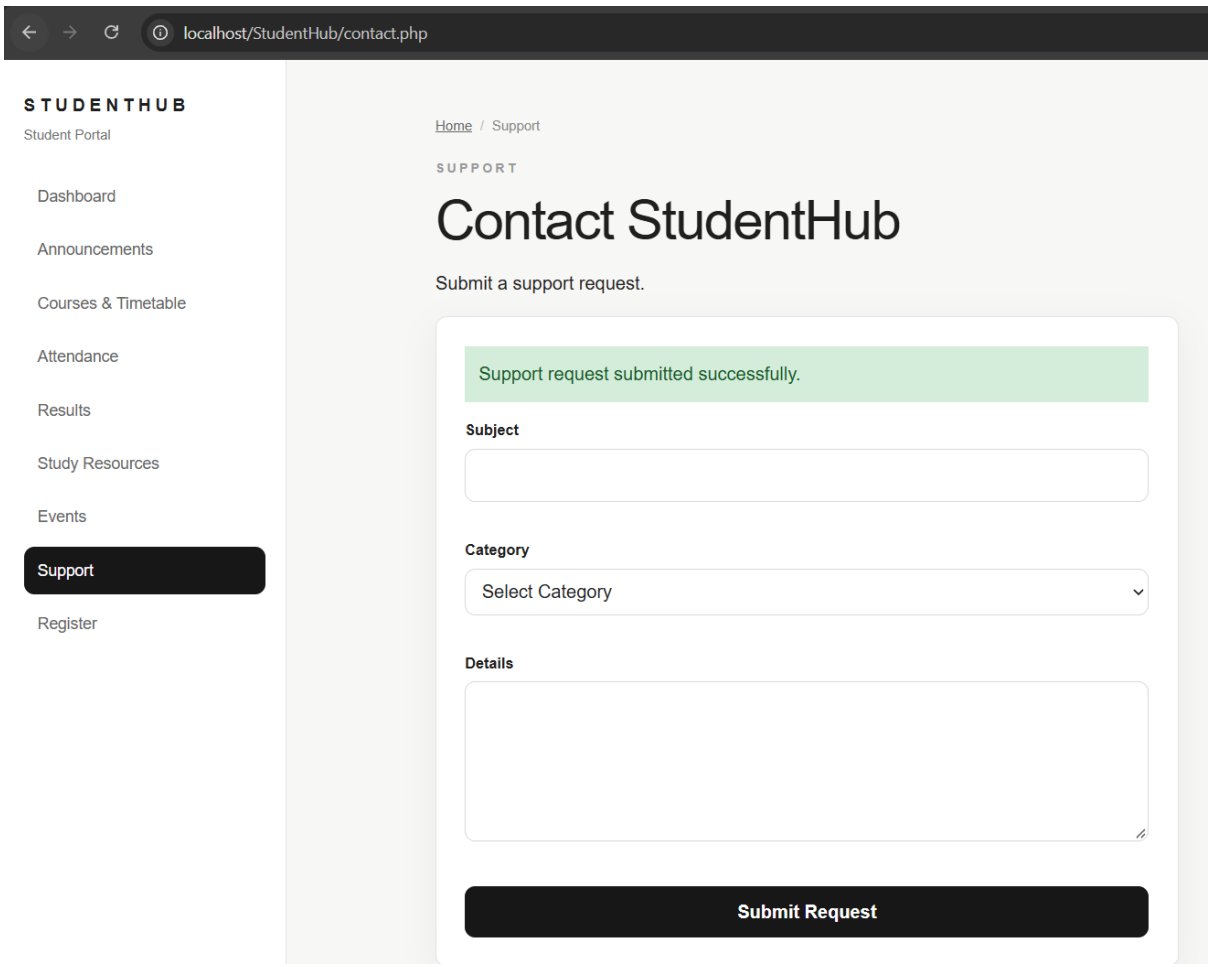
        if (file_put_contents($file, json_encode($contacts, JSON_PRETTY_PRINT), LOCK_EX) !== false) {
            $message = "Support request submitted successfully.";
            $messageType = "success";
        } else {
            $message = "Unable to save request.";
            $messageType = "error";
        }
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>StudentHub - Contact</title>
<link rel="stylesheet" href="styles.css">
```

```
<style>
.form-box{max-width:650px;padding:25px;background:#fff;border-radius:10px}
.form-group{margin-bottom:18px}.form-group label{display:block;margin-bottom:6px;font-weight:bold}
.form-group input,.form-group select,.form-group textarea{width:100%;padding:10px;box-sizing:border-box}
.form-group textarea{min-height:140px;resize:vertical}.form-box button{padding:10px 20px;cursor:pointer}
.success{padding:12px;margin-bottom:15px;background:#d4edda;color:#155724}
.error{padding:12px;margin-bottom:15px;background:#f8d7da;color:#721c24}
</style>
</head>
<body>
<div class="layout">
<aside class="sidebar">
<header><h2>STUDENTHUB</h2><p>Student Portal</p></header>
<nav aria-label="Student Portal Navigation">
<ul>
<li><a href="dashboard.html">Dashboard</a></li>
<li><a href="announcements.html">Announcements</a></li>
<li><a href="timetable.html">Courses &amp;amp; Timetable</a></li>
<li><a href="attendance.html">Attendance</a></li>
<li><a href="results.html">Results</a></li>
<li><a href="materials.html">Study Resources</a></li>
<li><a href="events.html">Events</a></li>
<li><a href="contact.php" aria-current="page">Support</a></li>
<li><a href="register.php">Register</a></li>
</ul>
</nav>
</aside>
<main class="content">
<nav aria-label="Breadcrumb"><ol><li><a href="dashboard.html">Home</a></li><li aria-current="page">Support</li></ol></nav>
<header><p>SUPPORT</p><h1>Contact StudentHub</h1><p>Submit a support request.</p></header>
<section class="form-box">
<?php if ($message !== ""): ?><div class="<?php echo $messageType; ?>"><?php echo $message; ?></div><?php endif; ?>
<form method="POST" action="contact.php">
<div class="form-group"><label for="subject">Subject</label><input type="text" id="subject" name="subject" required></div>
<div class="form-group"><label for="category">Category</label>
<select id="category" name="category" required>
<option value="">Select Category</option><option value="academic">Academic</option><option value="attendance">Attendance</option><option value="results">Results</option><option value="portal">Portal Access</o
</select></div>
<div class="form-group"><label for="details">Details</label><textarea id="details" name="details" required></textarea></div>
<button type="submit">Submit Request</button>
</form>
</section>
</main>
</div>
</body>
</html>
```

Output:




localhost/StudentHub/contact.php
STUDENTHUB
Student Portal
Dashboard
Announcements
Courses & Timetable
Attendance
Results
Study Resources
Events
Support
Register
Home / Support
SUPPORT
Contact StudentHub
Submit a support request.
Subject
web development
Category
Academic
Details
Moodle is not open
Submit Request


localhost/StudentHub/contact.php
STUDENTHUB
Student Portal
Dashboard
Announcements
Courses & Timetable
Attendance
Results
Study Resources
Events
Support
Register
Home / Support
SUPPORT
Contact StudentHub
Submit a support request.
Support request submitted successfully.
Subject
Category
Select Category
Details
Submit Request

Key Questions:

1. Is the form submitted using POST?

Ans: Yes. The form is submitted using the POST method. POST sends the form data to the PHP server without displaying the submitted values in the URL. PHP checks the request using $_SERVER["REQUEST_METHOD"] and processes the data using $_POST.

Q2. Are inputs validated and sanitized server side?

Ans: Yes. All form inputs are validated and sanitized on the server side using PHP. Required fields are checked, email addresses are validated using filter_var(), and mobile numbers are checked using regular expressions. htmlspecialchars() is used to sanitize input and reduce the risk of malicious HTML/script content.

3. Is file writing handled safely?

Ans: Yes. The submitted data is stored safely in **CSV and JSON files**. CSV data is written using fputcsv(), while JSON data is stored using json_encode(). The JSON file uses LOCK_EX during writing to reduce the chance of simultaneous write conflicts. Files are created when they do not already exist.

4. Are success and error responses displayed clearly?

Ans: Yes. The PHP program displays clear success and error messages after form submission. For example, a valid submission displays a success message such as "Registration successful!", while invalid or incomplete data displays an appropriate error message such as "Please enter a valid email address." This helps users understand whether their submission was successful or needs correction.

GitHub Link: https://github.com/smit-2406/studenthub-website/tree/main/WD